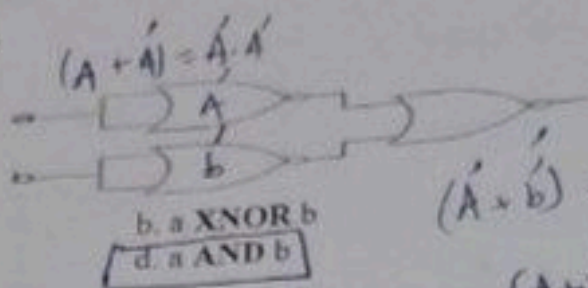


7. The result of the following circuit is :



- a. a XOR b
c. a OR b

8. The output of the following Boolean expression after simplification.

$$F = (A+B)(A'B') + A\bar{A}B + \bar{A}\bar{B}A + BAB$$

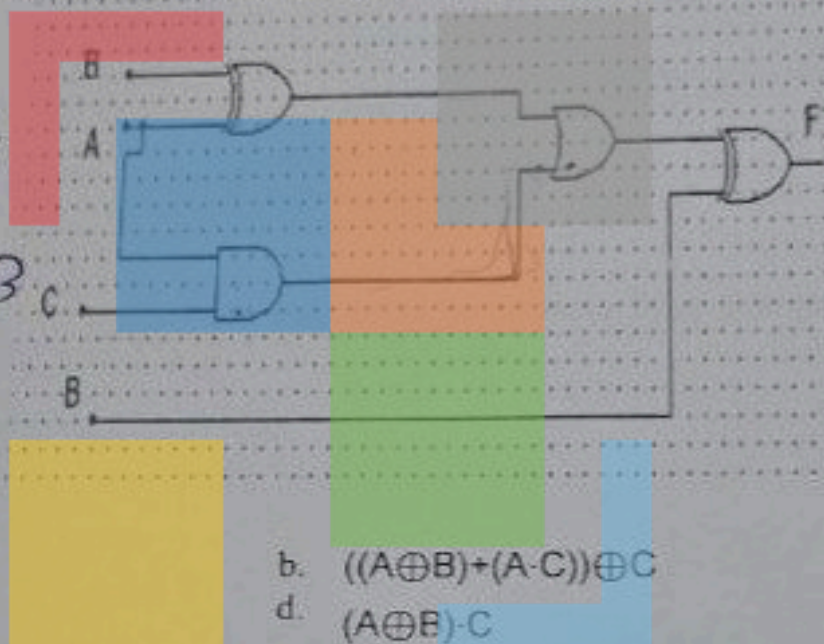
- a. $A\bar{B}$
c. 1

b. 0
d. $(A \oplus B)$

9. Which Boolean equation represents the output of the circuit?

~~PSF~~
 $(B\bar{A} + \bar{B}A) \cdot (AC)$

$(A \oplus B - AC) \oplus B$



$((A \oplus B) + (A \cdot C)) \oplus B$
 $(A \cdot C) + (A \oplus B)$

- b. $((A \oplus B) + (A \cdot C)) \oplus C$
d. $(A \oplus B) \cdot C$

The simplification of the Boolean expression is.

$$\overline{A}B\bar{C} + A\bar{B}C = A + B + C' + \bar{A} + B + \bar{C}$$

- b. 0
d. BC

Boolean expression $\bar{A}.B + A.\bar{B} + A.B$ is equivalent to

$B(\bar{A} + A) + A.B$

$B + A.B$

$(B + B)A + B$

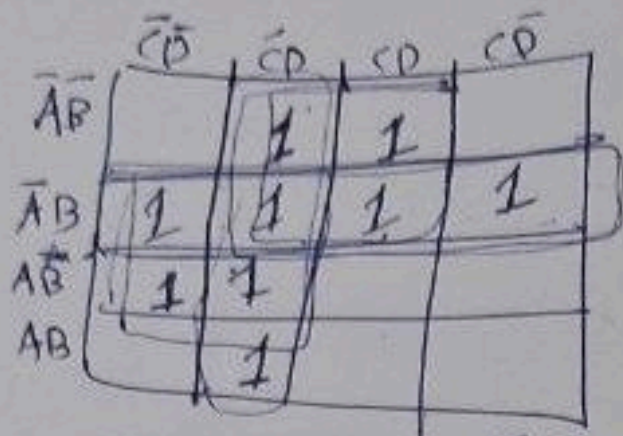
- b. $\bar{A}.B$
d. $A.B$

Output of a logic gate is 1 when all its inputs are at logic 0, the gate is either

- OR or an EX-OR
NOR or an EX-NOR

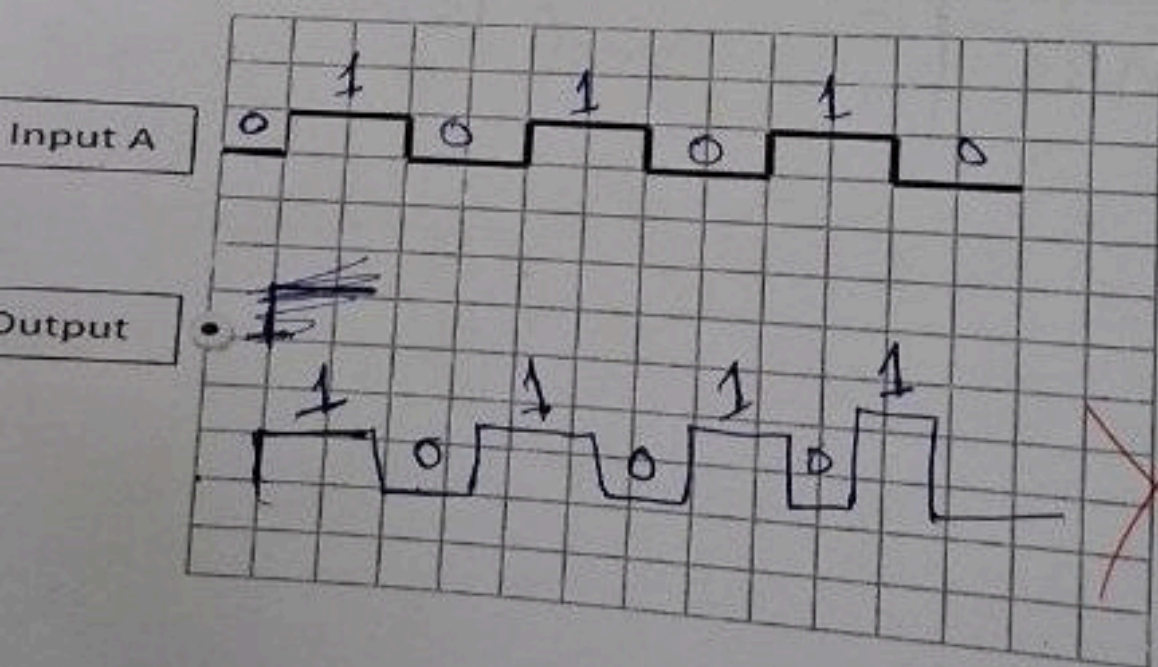
- b. an OR or an EX-NOR
d. a NOR or an EX-NOR

Q2) Simplify the following expression into sum of products using Karnaugh map
 $F(A,B,C,D) = \sum(1,3,4,5,6,7,9,12,13)$ (6marks)



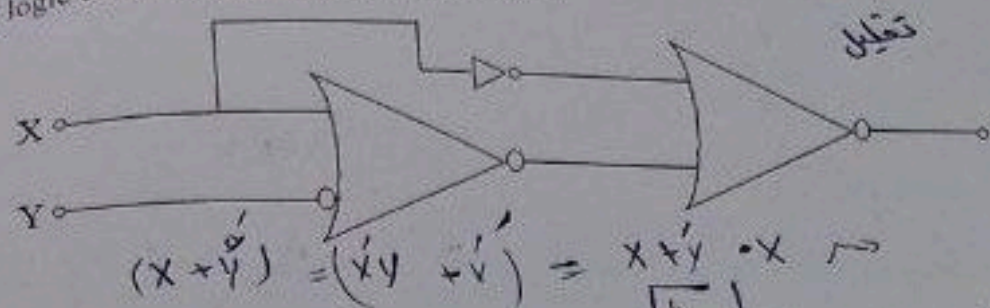
$P = \bar{A}B + \bar{C}B + \bar{A}D + \bar{C}D$ equal

following diagram shows input waveform for a two-input XNOR gate, inputs were connect
 e output waveform of this function? (3marks)



XNOR

8. The logic circuit shown can be minimized to

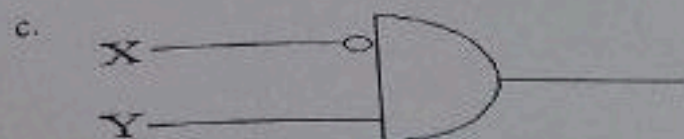


$$(X + Y') = (Y' + X) = X + Y' = X$$

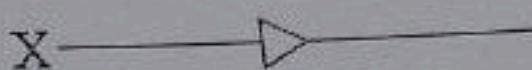
$$X + Y'X = X + X(Y' + X)$$

a. X Inverter Not

b.



d.



14. Which of following are known as universal gates

a. NAND & NOR
c. XOR & OR.

b. AND & OR
d. None.

15. How many two-input AND and OR gates are required to realize $Y = CD + EF + G$

a. 2, 3
c. 2, 1.

b. 2, 2.
d. 3, 3

Fill the answers for Q1 here

1	2	3	4	5	6	7	8	9	10	11	12	13
c	c	c	b	c	c	d	b	a	c	a	d	
✓	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗

D

12

Select the correct answer, from options (a, b, c, d) to the following questions. (15 marks)

Which of the following statements best describes the XOR operation?

- XOR returns 1 if both inputs are 1.
XOR returns 1 if only one of the inputs is 1.

- b. XOR returns 1 if both inputs are 0.
d. XOR returns 1 if both inputs are the same.

If you have a 3-input AND gate but want to use only 2 inputs, how can you handle the unused input?

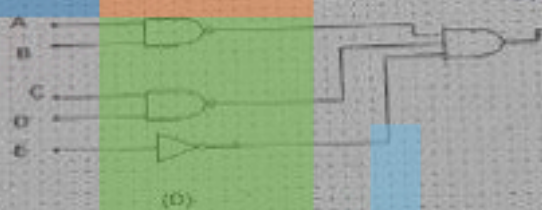
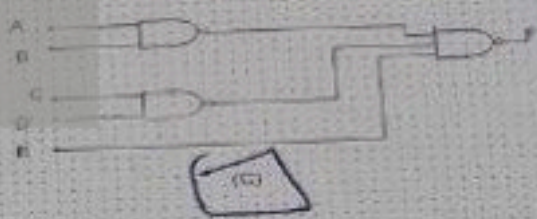
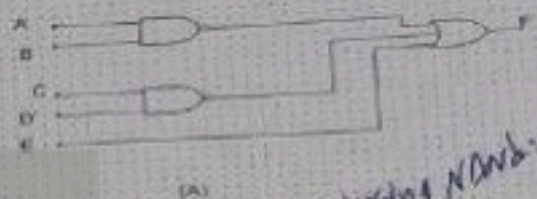
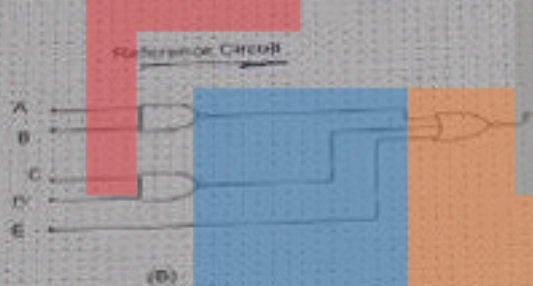
Leave the unused input disconnected.

Connect the unused input to logic HIGH (1).

b. Connect the unused input to logic LOW (0).

d. It doesn't matter; the result will be the same regardless of the unused input's connection.

Which of the figures (A to D) is equivalent to the reference circuit.



- b. (B)
d. (D)

Given a 3-variable function $F(A,B,C) = \sum(1,3,5,7)$, which of the following expressions represents the function in terms of minterms?

$F = A'B'C + A'BC + AB'C + ABC$
 $F = A'B'C + AB'C + ABC + A'BC$

b. $F = A'B'C + A'BC + AB'C + ABC$
d. $F = A'BC + A'B'C + AB'C + ABC$

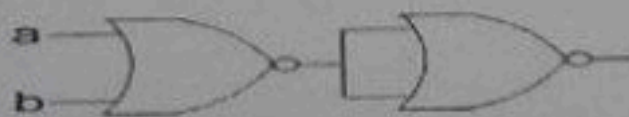
Which of the following statements is true?

$(A + B)(A + C) = AC + BC$
 $(A + B)(A + C) = A + BC$

b. $(A + B)(A + C) = AB + C$
d. $(A + B)(A + C) = AC + B$

The result of the following circuit is:

$(a \cdot b)' = a'$



$(a+b)' =$

XOR b
OR b

- b. a XNOR b
d. a AND b

$(a+b)' = (a'b)' = a+b$